# 中国船级社
# CHINA CLASSIFICATION SOCIETY
# 船用产品证书
# CERTIFICATE OF MARINE PRODUCT

证书编号/Certificate No.
**ZG26PPS01016**

**兹证明**本证书所列产品经本社署名验船师检验，符合本证书注明标准的要求。
**This is to certify that** the following products have been inspected by the undersigned surveyor of the Society and are found to comply with the requirements of the specified standards.

| | |
|---|---|
| **产品名称 Product** | **铜合金螺旋桨 Copper Alloy Propeller** |
| 申请方 Applicant | 浙江元鼎船舶设备有限公司 Zhoushan East Ocean Propeller-Making Co.,Ltd. |
| 制造厂 Manufacturer | 浙江元鼎船舶设备有限公司 Zhoushan East Ocean Propeller-Making Co.,Ltd. |
| 订货方 Purchaser | 杭州前进齿轮箱集团股份有限公司 Barge Hengue 009 |

认可证书号/Certificate No. of Approval ZG26PWA00001 图纸批准号/Approval No. of Drawings 详见附页/See additional page(s)

用于/Intended for 国际航行海船/International sea-going ship
船舶工程号/Hull No.:VHX9150

产品编号/Serial No. **详见附页/See additional page(s)**

## 产品检验标准/Product Inspection Standard

1.中国船级社《材料与焊接规范》（2025）及其变更通告第1篇第9章
Chapter 9, Part One of China Classification Society Rules for Materials and Welding 2025 and its Change Notices

## 产品明细/Product Description

**铜合金螺旋桨/Copper Alloy Propeller**(M0001)

| 名称/Name | 属性（值）/Value | 单位/Unit |
|---|---|---|
| 结构/Structure | 详见附页/See additional page(s) | |
| 合金类型/Alloy Types | 详见附页/See additional page(s) | |
| 数量/Quantity | 详见附页/See additional page(s) | |
| 炉号/Cast No. | 详见附页/See additional page(s) | |
| 无损检测/Non-Destructive Test | 详见附页/See additional page(s) | |
| 图号/Drawing No. | 详见附页/See additional page(s) | |
| 直径/Diameter | 详见附页/See additional page(s) | |
| 叶片数量/Number of Blade | 详见附页/See additional page(s) | |
| 旋转方向/Direction of Rotation | 详见附页/See additional page(s) | |
| 重量/Weight | 详见附页/See additional page(s) | |

## 产品检验标志/Marking

位置/Position : 在产品上 / On the product
标志样式 Type of Marking : ZG26PPS01016 钢印标志 / CCS steel stamp

发证日期 Date of issue 2026年02月27日 Feb. 27, 2026

验船师 Surveyor Wang Pingzong

本证书根据中国船级社规范和相关规定签发。所有证书页为一个整体，必须同时使用。纸质证书每页均须由本社盖章方为有效，电子证书含数字签名方为有效，本证书复印件无效。任何单位和个人均不应摘录或节选本证书的部分内容。有关方对所持证书的真实性有疑问时，可以向我社检验机构咨询。
This Certificate is issued pursuant to the Rules of the Society and related regulation. All pages of the certificate are taken as a whole and are used simultaneously. No paper certificate page is valid without bearing the stamp of the Society, no electronic certificates is valid without the digital signature, and no copied form of the certificate is regarded as valid. Any part of the certificate is not to be extracted or abridged by any unit or individual in any form. Related parties who are doubted about the authenticity of the certificate may inquire of the Society or its offices.

## 备注/Remarks

无/Nil.

**中国船级社浙江分社**
**CCS Zhejiang Branch**
注：本证书含有附页，共2页
Note: The certificate is attached with additional 2 page(s)

产品明细/Product Description
铜合金螺旋桨/Copper Alloy Propeller（M0001）

| 名称/Name | 属性（值）/Value | 单位/Unit |
|---|---|---|
| 结构/Structure | 整体式/Integral type | |
| 数量/Quantity | 1 | 件/Pcs |
| 产品编号/Serial No. | DH26010022-1 | |
| 叶片数量/Number of Blade | 4 | 片/Pcs |
| 重量/Weight | 761 | Kg |
| 图号/Drawing No. | HS9266-425-016 | |
| 图纸批准号/Approval No. of Drawings | WP26SNP00026-002H | |
| 表面状态/Surface Machining | 精加工/Finished machining | |
| 无损检测/Non-Destructive Test | 着色探伤/PT | |
| 炉号/Cast No. | 2405161 | |
| 合金类型/Alloy Types | CU3 | |
| 0.7R 处螺距/Pitch at 0.7r | 1720 | mm |
| 侧斜角/Side Bevel | 5.44 | 度/degree |
| 静平衡实验/Static Balance Test | 合格/Qualified | |
| 直径/Diameter | 2450 | mm |
| 旋转方向/Direction of Rotation | 左旋/Left ration | |
| 断后伸长率/Percentage Elongation After Fracture（A5） | 20% | |
| 抗拉强度/Tensile Strength（Rm） | 620 | MPa |
| 规定非比例延伸强度/Proof Strength（Rp0.2） | 271 | MPa |
| 化学成分/Chemical Composition | Cu：80.01%, Al：8.92% , Mn：1.03% , Zn：0.35% , Fe：4.97% , Ni：4.09%, Sn：0.07%, Pb：0.03% , Si：0.056% | |

中国船级社浙江分社 CCS Zhejiang Branch

铜合金螺旋桨/Copper Alloy Propeller（M0002）

| 名称/Name | 属性（值）/Value | 单位/Unit |
|---|---|---|
| 结构/Structure | 整体式/Integral type | |
| 数量/Quantity | 1 | 件/Pcs |
| 产品编号/Serial No. | DH26010022-2 | |
| 叶片数量/Number of Blade | 4 | 片/Pcs |
| 重量/Weight | 763 | Kg |
| 图号/Drawing No. | HS9266-425-016 | |
| 图纸批准号/Approval No. of Drawings | WP23SNP00026-002H | |
| 表面状态/Surface Machining | 精加工/Finished machining | |
| 无损检测/Non-Destructive Test | 着色探伤/PT | |
| 炉号/Cast No. | 2405154 | |
| 合金类型/Alloy Types | CU3 | |
| 0.7R 处螺距/Pitch at 0.7r | 1720 | mm |
| 侧斜角/Side Bevel | 5.44 | 度/degree |
| 静平衡实验/Static Balance Test | 合格/Qualified | |
| 直径/Diameter | 2450 | mm |
| 旋转方向/Direction of Rotation | 右旋/Right ration | |
| 断后伸长率/Percentage Elongation After Fracture（A5） | 20% | |
| 抗拉强度/Tensile Strength（Rm） | 620 | MPa |
| 规定非比例延伸强度/Proof Strength（Rp0.2） | 273 | MPa |
| 化学成分/Chemical Composition | Cu：80.01%, Al：8.92% , Mn：1.03% , Zn：0.35% , Fe：4.97% , Ni：4.09%, Sn：0.07%, Pb：0.03% , Si：0.057% | |

中国船级社浙江分社 CCS Zhejiang Branch